Designed to work beyond the screen.

SYSTEM SPECIFICATION / FLUTTER + LARAVEL

Version 1 · explicit decisions, contracts and acceptance criteria · static design / implementation handoff

Architecture & product contract

An offline-first personal habit system. This specification defines intended behavior; the accompanying Figma screens are not a functioning application.

Runtime and responsibility

Target from the brief: Flutter 3.44.x mobile client; PHP 8.4 with Laravel 12 API and Sanctum bearer-token authentication. Confirm supported versions during implementation. PostgreSQL is the default transactional database. Flutter uses drift / SQLite as its working copy, so a check-in never waits for the network.

Data path

```
Tap → one local SQLite transaction (projection + durable outbox)
Outbox → POST /api/v1/sync → Laravel owner-scoped transaction
SQL state + reward/freeze ledgers + monotonically ordered change journal
Server delta → durable local apply → advance server cursor → update derived UI
```

Ownership boundaries

Field / resource	Type / method	Contract / constraint
Flutter + drift	Local working copy	Immediate logs, outbox, optimistic progress; confirmed server streak / XP kept separately.
Laravel + Sanctum	Authoritative service	Authentication, ownership, dates, validation, streaks, XP, freeze transactions, versioned APIs.
SQL + worker queue	Durable server state	Row locks, unique constraints, change sequence, export jobs, milestone push outbox.
Device notifications	Local scheduling	Per-habit reminders without network dependence; permissions and OS restrictions still apply.
FCM / APNs	Separate push route	Server-triggered milestone messages only; not the dependency for daily reminders.

Decision baseline

Field / resource	Type / method	Contract / constraint
Three habit types	binary / quantity / duration	Binary 0 or 1; exact-decimal quantity; whole duration seconds. A daily row accumulates increments.
Four schedules	daily / weekdays / weekly_count / interval	Monday-start local weeks; interval anchored to start date. Weekly target means distinct completed local days.
Gentle rewards	10 XP / completed habit-day	No paid economy, leaderboards, guild damage, coercive copy or random reward promises.
Freezes	Opt-in; shared cap 2	One missed closed scheduled day or failed week costs one token. Monthly replenishment tops up to cap.
Example account	Maya Chen · Pacific Time	Today 3/5, then meditation makes 4/5. XP 1,240 → 1,250; Level 13, 50/100 toward 14.
Visible state contract	24 mobile screens	Access/setup; Today/logging; CRUD/schedule/reminders; heatmap/insights; XP/freezes; sync conflicts; privacy/archive; first-run/error.

Rollout without changing the core contract

MVP: authentication, binary habits, daily logs, offline outbox and sync, heatmap/streaks, local reminders. Advanced: quantity/duration, weekly/interval schedules, versioned edits, XP/freezes, category insights, exports and restore jobs. The design shows the advanced scope. Background delivery, imports and milestones are independently testable releases.

Schema · identity & habit definitions

UUID identifiers throughout; owner foreign keys on all user resources. All event timestamps are timestamptz in UTC. Business dates are DATE, not converted UTC timestamps.

users

Field / resource	Type / method	Contract / constraint
id / name / email	uuid PK / varchar / varchar	Email unique, normalized; display name max 80 characters.
password	varchar(255)	Laravel password hash, never returned or exported.
timezone	varchar(64)	Validated IANA zone, default selected during onboarding.
xp / freeze_balance	bigint / smallint	Cached sums of ledgers; xp ≥ 0, freeze balance between 0 and 2. Clients cannot submit either.
auto_freeze / timezone_mode	boolean / enum	Auto-freeze opt-in; timezone_mode fixed or ask_on_device_change.
created_at / updated_at / deleted_at	UTC timestamps	Soft account deletion only while purge policy / owned-resource cleanup runs.

habits

Field / resource	Type / method	Contract / constraint
id / user_id / name	uuid PK / uuid FK / varchar(100)	Index user_id + archived_at; every query owner-scoped.
type / unit / category	enums / varchar(24)	Type binary, quantity, duration. Category health, mindfulness, learning, productivity, other.
target_value	numeric(12,3)	Quantity exact decimal > 0; binary fixed 1; duration whole seconds > 0. Max documented and validated.
frequency_type	enum	daily, weekdays, weekly_count or interval.
frequency_config	jsonb	daily {}; weekdays {days:[1..7]}; weekly_count {count:1..7}; interval {every_n_days:1..365, anchor_date:DATE}.
start_local_date / archived_at	DATE / UTC nullable	No eligibility before start. Archive creates an active-range end, not retroactive failures.
version / definition_version	bigint / bigint	Optimistic concurrency for metadata; refers to the effective schedule / target revision.
created_at / updated_at	UTC timestamps	Do not use these timestamps as the sync cursor.

Versioned schedule and timezone support

Field / resource	Type / method	Contract / constraint
habit_definition_versions	id UUID; habit_id; effective_date DATE; type; target; unit; category; frequency JSON	UNIQUE(habit_id,effective_date). New definitions start next eligible period; never reevaluate older logs with a new target.
habit_active_ranges	id UUID; habit_id; starts_on DATE; ends_before DATE nullable	Half-open active local-date range; non-overlapping per habit. Restore creates a new range from today.
user_timezone_history	id UUID; user_id; timezone IANA; effective_at UTC; created_at UTC	Ordered effective timeline. A validated event uses the zone effective at occurred_at, not the current profile zone.
habit_streak_cache	habit_id PK; current; longest; unit; computed_through; dirty_from; version	Derived cache only. Unit is days / scheduled periods or weeks, based on frequency. Invalidated transactionally after log / definition changes.

Historical invariants

Freeze usage, log completion and rewards refer to effective definition versions and period keys. Historical timezone changes do not rewrite log_date or old period boundaries. Active ranges survive archive/restore. Index histories by (habit_id, effective_date) and (user_id, effective_at). Foreign-key deletion policy follows explicit account purge, never accidental cascade from an archive.

Schema · logs, reminders & devices

The normalized daily log is a projection; idempotent immutable mutations are the audit trail. A unique habit/day row prevents duplicate cups, sessions or completion awards.

habit_logs

Field / resource	Type / method	Contract / constraint
id / habit_id / log_date	uuid PK / uuid FK / DATE	UNIQUE(habit_id,log_date). Index (habit_id,log_date DESC). No duplicate row for another sip.
value	numeric(12,3)	Binary {0,1}; quantities exact decimal; duration integer seconds. Non-negative validated type-specific bounds.
occurred_at / completed_at	UTC / UTC nullable	Latest contributing occurrence; completed_at set when effective target is reached, cleared when reversed.
resolved_timezone / definition_version	IANA text / bigint FK	Frozen resolution context; target revision used for that business date.
version	bigint	Monotonic on mutation; used by absolute edits and deletes. Increments use receipt dedupe plus atomic addition.
deleted_at / created_at / updated_at	UTC timestamps	Soft tombstone. Reactivation of the same day keeps the stable log identity and increments version.

reminders

Field / resource	Type / method	Contract / constraint
id / habit_id	uuid PK / uuid FK	Owner inferred through habit; archive disables device scheduling.
local_time / days_of_week	time / jsonb array	Clock time, not UTC time; unique days 1=Mon…7=Sun. Never silently apply server timezone.
timezone_mode / timezone	enum / IANA nullable	habit_zone or device_zone; explicit mode is shared, OS scheduling remains local per device.
enabled / version	boolean / bigint	Definition enabled is separate from device permission being denied. Versioned updates.
created_at / updated_at / deleted_at	UTC timestamps	Journal all definition changes and deletions.

devices

Field / resource	Type / method	Contract / constraint
id / user_id / platform	uuid PK / uuid FK / enum	platform ios or android. Device records owner-scoped.
push_token / token_updated_at	encrypted text nullable / UTC	FCM registration token or routed APNs token. Restricted storage and logs; remove invalid tokens.
notification_permission / device_timezone	enum / IANA	unknown, authorized, provisional, denied; diagnostics, not a promise of delivery.
last_seen_at / revoked_at	UTC timestamps	Separate from Sanctum token expiry. Revoke both on logout for that device.
created_at / updated_at	UTC timestamps	Unique active provider token. No token in public user or export payloads.

Useful derived state

Field / resource	Type / method	Contract / constraint
completion	value ≥ effective target	Binary 1; quantity threshold; duration seconds threshold. Derived server-side; clients only show provisional completion offline.
weekly progress	COUNT distinct completed log_date	Only active local days inside the Monday–Sunday period count; excess sessions same day do not inflate target count.
consistency	completed closed eligible periods / all closed eligible periods	Exclude open/future/not-due/archived dates. Frozen periods are neutral visually but not numerator successes.
local SQLite projection	Same resource UUIDs + server version	Store local pending flags separately. Never overwrite an unsent absolute edit with a pulled server snapshot without preserving review data.

Schema · durable sync & reward accounting

All balance-changing work is transactional, auditable and idempotent. Cached XP and freeze balances can be reconstructed; client values are never authoritative.

Mutation and change journal

Field / resource	Type / method	Contract / constraint
mutation_receipts	user_id UUID; mutation_id UUID; device_id UUID; operation text; payload_hash text; result JSON; committed_at UTC	UNIQUE(user_id,mutation_id). Same key + changed payload is a 409 idempotency mismatch. Store durable outcome and canonical resolved date.
server_changes	seq bigint identity; user_id UUID; entity_type; entity_id UUID; operation upsert/delete; version bigint; payload JSON; created_at UTC	Index (user_id,seq). Every committed change, including deletion, has a sequence. Cursor is opaque and bound to user.
local_outbox (drift)	mutation_id UUID PK; payload JSON; base_version; created_at; attempts; state; server_result	states queued, sending, accepted, needs_review, rejected. Persist before optimistic UI; never purge unacknowledged mutations.

Reward ledgers

Field / resource	Type / method	Contract / constraint
xp_entitlements	id UUID; user_id; habit_id; log_date DATE; amount=10; state active/reversed; revision	UNIQUE(habit_id,log_date). Stable entitlement caps one net reward per completed habit-day. Not recreated after deletion.
xp_ledger	id UUID; user_id; entitlement_id; entitlement_revision; delta bigint; reason; mutation_id; created_at UTC	UNIQUE(entitlement_id,entitlement_revision). Completion +10, reversal −10, recompletion +10 only after a prior reversal. Append-only.
freeze_ledger	id UUID; user_id; kind grant/ spend/refund; delta smallint; reason; source_key; created_at UTC	UNIQUE(user_id,source_key). Balance sum between 0–2 under user wallet lock; zero-delta capped refunds are audit events.
freeze_usage	id UUID; user_id; habit_id; period_key; definition_version; used_on DATE; spend_ledger_id; reconciled_at UTC nullable; state active/refunded	UNIQUE(habit_id,period_key). Exactly one spend per eligible closed period; at most one reconciliation/refund.
period_evaluations	habit_id; period_key; definition_version; start_date; end_date; timezone; starts_at UTC; ends_at UTC; completed; protected; revision	UNIQUE(habit_id,period_key). Persistent closure ordering and reevaluation watermark, including late offline contributions.

Data portability jobs

Field / resource	Type / method	Contract / constraint
export_jobs	id UUID; user_id; format csv/json; status queued/running/ready/ failed/expired; snapshot_seq; object_key; expires_at; error_code; timestamps	Consistent server snapshot; private object, signed owner-only URL with 24h expiry. Do not export secrets or push tokens.
import_jobs	id UUID; user_id; file_hash; schema_version; status validating/review/queued/ running/complete/failed; plan JSON; timestamps	Preview and conflict review before apply. A verified own-export manifest can restore older history; untrusted CSV follows ordinary backdate limits.
push_outbox	id UUID; user_id; device_id; event_key unique; payload JSON; status; attempts; next_attempt_at UTC	Transactional milestone enqueue; dedupe independently of local reminder notifications.

Recommended migration order

1 users/timezone history + Sanctum tokens; 2 habits/definitions/active ranges; 3 logs/streak cache/period evaluations; 4 reminders/devices; 5 receipts/change journal/outbox bookkeeping; 6 XP entitlements/ledger; 7 freeze ledger/usage; 8 export/import/push jobs. Add checks and indexes in the same migration as each table. Use migration tests and transaction rollback tests, not only application validation.

API · authentication & habit lifecycle

Base path /api/v1. JSON over HTTPS. Bearer token required except auth and reset routes. Never accept owner IDs, XP, level, streak or freeze balances from clients.

Authentication and identity routes

Field / resource	Type / method	Contract / constraint
POST /auth/register	201	name, email, password, timezone, device_name. Create account + timezone history + scoped Sanctum token.
POST /auth/login	200	email, password, device_name. Rate-limited; return token once, user and server_time.
POST /auth/logout	204	Revoke current token and its device registration. /auth/logout-all revokes all user tokens and devices.
POST /auth/forgot-password	202	email; uniform response avoids email enumeration.
POST /auth/reset-password	204	email, reset_token, password, password_confirmation; one-time expiring reset, revoke old sessions.
GET /me · PATCH /me	200	Readable profile; editable name and validated future-effective timezone. Capture timezone timeline.
DELETE /me	202	Recent reauthentication required. Revoke sessions immediately, queue owned-data purge, expose job status.

Register request and response

```
POST /api/v1/auth/register
{
  "name": "Maya Chen",
  "email": "maya.chen@example.com",
  "password": "12-plus-character-password",
  "timezone": "America/Los_Angeles",
  "device_name": "Maya's iPhone"
}
201 {
  "data": {
    "user": {
      "id": "11111111-1111-4111-8111-111111111111",
      "name": "Maya Chen",
      "timezone": "America/Los_Angeles"
    },
    "token": "RETURN_ONCE",
    "token_type": "Bearer"
  },
  "meta": {
    "api_version": "v1",
    "server_time": "2026-05-28T17:22:00Z"
  }
}
```

Habit resource routes

Field / resource	Type / method	Contract / constraint
GET /habits	200; cursor pagination	active or archived filter, category, q; default limit 50, max 100. Response data[] plus next_cursor.
POST /habits	201 + Location	name, type, target_value, unit, category, frequency_type/config, start_local_date. UUID supplied for offline creation or generated server-side.
GET /habits/{id}	200	Canonical definition, active ranges, computed streaks/period progress and version. Return owner-filtered 404 if not found.
PATCH /habits/{id}	200 / 409	base_version required. Target/frequency revisions apply next eligible period; return effective_date and definition_version.
POST /habits/{id}/archive	200	base_version + archived effective date (server resolves today); no history deletion.
POST /habits/{id}/restore	200	base_version; new active range from server-resolved local today. Previous archived intervals remain excluded.

Create a quantity habit

```
POST /api/v1/habits
{
  "id": "22222222-2222-4222-8222-222222222222",
  "name": "Water",
  "type": "quantity",
  "target_value": "2000.000",
  "unit": "mL",
  "category": "health",
  "frequency_type": "daily",
  "frequency_config": {},
  "start_local_date": "2026-05-01",
  "date_mode": "backdate"
}
201 {
  "data": {
    "id": "22222222-2222-4222-8222-222222222222",
    "version": 1,
    "definition_version": 1,
    "archived_at": null
  },
  "meta": {
    "server_time": "2026-05-28T17:22:00Z"
  }
}
```

Validation, identity and concurrency

Password minimum 12 characters. Name max 100; type/unit/target compatible; weekday list unique; weekly target 1–7; interval 1–365. Future definition effective date comes from server schedule boundaries. Clients persist base_version with edits; a 409 includes canonical current resource and version. Use the same mutation application service for online mutations and /sync to avoid divergent business rules. Past start dates require date_mode=backdate within30 days.

API · check-ins, insights & portability

Domain writes share the same mutation envelope and service as offline sync. Idempotency-Key equals mutation_id; exact decimals travel as strings, durations as whole seconds. Examples are independent fixtures unless stated otherwise.

Log and read contracts

Resource / scenario	Method / rule	Contract / expected behavior
POST /habits/{id}/logs	200 / 201	operation set_binary, increment_quantity, increment_duration or set_value. mutation_id, occurred_at, timezone context; base_version for absolute set.
PATCH /logs/{id}	200 / 409	Absolute value edit: base_version + mutation_id. Recompute completion, XP entitlement and streak periods.
DELETE /logs/{id}	200	base_version + mutation_id. Return tombstone, new version and confirmed reward delta. No physical row deletion.
POST /logs/{id}/restore	200 / 409	Reviewed restoration with tombstone base_version and value. Stale queued increments must not resurrect deleted logs.
GET /habits/{id}/logs	200; paginated	from/to local DATE, limit50 max100, cursor. Read includes effective target, value, completion, version and timezone.
GET /habits/{id}/heatmap	200	from/to DATE ≤366-day window; complete/protected/missed/not_due/ pending status and legend.
GET /insights	200	period week/month, local window, category filter; exact completed/due counts plus rounded percent.
GET /rewards · GET /freezes	200; paginated ledger	Server XP/level and freeze balance. PATCH /me/settings toggles auto_freeze with effective policy history.

Reminder, device and portability routes

Resource / scenario	Method / rule	Contract / expected behavior
GET /reminders · POST / reminders · PATCH /reminders/{id} · DELETE /reminders/{id}	200 / 201 / 204	Owner habit, local_time, days_of_week, timezone_mode, enabled; base_version on update/delete. Definition syncs; permission is device-specific.
PUT /devices/{id} · DELETE / devices/{id}	200 / 204	Register/revoke owned device/platform/token/permission. Another owner's device ID returns404.
POST /exports · GET /exports/{id}	202 / 200	format json/csv. State queued/running/ready/failed/expired; ready response has private download URL and expires_at.
POST /imports · GET /imports/{id} · POST /imports/{id}/confirm	202 / 200 / 202	Upload, validate ownership/hash/schema, preview conflicts, confirm plan. Never restore claimed XP/freezes directly.
GET /account-deletions/{id}	200	Deletion status only, via scoped status receipt after full tokens are revoked.

Quantity increment and canonical response

```
POST /api/v1/habits/22222222-2222-4222-8222-222222222222/logs
Idempotency-Key:33333333-3333-4333-8333-333333333333
{
  "mutation_id": "33333333-3333-4333-8333-333333333333",
  "operation": "increment_quantity",
  "value": "250.000",
  "occurred_at": "2026-05-28T17:15:00Z",
  "captured_timezone": "America/Los_Angeles",
  "local_date_hint": "2026-05-28"
}
200 {
  "data": {
    "log_date": "2026-05-28",
    "value": "1500.000",
    "unit": "mL",
    "target_value": "2000.000",
    "completed_at": null,
    "version": 9
  },
  "meta": {
    "duplicate": false,
    "reward_delta": 0
  }
}
```

Structured validation and conflict

```
422 {
  "error": {
    "code": "validation_failed",
    "message": "Check the highlighted fields.",
    "fields": {
      "target_value": [
        "Duration must be greater than zero."
      ]
    }
  },
  "meta": {
    "request_id": "REQUEST_UUID"
  }
}
409 {
  "error": {
    "code": "version_conflict",
    "resource_id": "LOG_UUID",
    "expected_version": 8,
    "current_version": 9,
    "current": {
      "value": "1750.000"
    }
  },
  "meta": {
    "request_id": "REQUEST_UUID"
  }
}
```

HTTP and validation baseline

401 missing/revoked token; owner-scoped404 for absence;403 authenticated action blocked by policy;409 stale version/idempotency mismatch/deleted resource;410 expired sync cursor;413 oversized batch;422 field/date/timezone validation;429 with Retry-After;5xx retriable failure. All responses carry request_id and server_time; resource responses also api_version. Examples omit repetitive envelope fields for brevity. Never expose secrets or another owner's data.

Sync · bounded delta protocol

Server-issued ordered cursors, immutable idempotent mutations and a durable local outbox. No full-table overwrites or client last_synced_at clocks.

Mutation semantics

Resource / scenario	Method / rule	Contract / expected behavior
log.increment_quantity / increment_duration	Atomic additive merge	Receipt applies once. Decimal-safe sum / integer seconds; threshold and XP recalculated transactionally. Concurrent increments commute.
log.set_binary / log.set_value / edit / delete / restore	Version-checked	Base version required;0 means absent. Identical desired state may ack a no-op; stale opposite intent conflicts, never clock-based last-write-wins.
habit.create / edit / archive / restore; reminder.*	Owner + version validation	Offline creation uses stable UUID. Apply habit before dependent log; dependency_pending is retriable.
Delete vs unsent increment	No silent resurrection	409 resource_deleted with tombstone version. Preserve local change for reviewed restore/discard.
Idempotency	UUID + canonical payload hash	Retry immutable original payload; changed intent gets new UUID. Durable receipts prevent replay after cursor reset.

POST /api/v1/sync · request

```
{
  "device_id": "44444444-4444-4444-8444-444444444444",
  "cursor": "OPAQUE_USER_CURSOR",
  "pull_limit": 200,
  "mutations": [
    {
      "mutation_id": "33333333-3333-4333-8333-333333333333",
      "entity": "habit_log",
      "entity_id": "55555555-5555-4555-8555-555555555555",
      "operation": "log.increment_quantity",
      "base_version": null,
      "occurred_at": "2026-05-28T17:15:00Z",
      "captured_timezone": "America/Los_Angeles",
      "local_date_hint": "2026-05-28",
      "payload": {
        "value": "250.000",
        "unit": "mL",
        "habit_id": "22222222-2222-4222-8222-222222222222"
      }
    }
  ]
}
```

Response · per-mutation ack plus bounded pull

```
200 {
  "data": {
    "acks": [
      {
        "mutation_id": "33333333-3333-4333-8333-333333333333",
        "status": "accepted",
        "duplicate": false,
        "resolved_date": "2026-05-28",
        "version": 9
      }
    ],
    "changes": [
      {
        "seq": 1051,
        "entity": "habit_log",
        "id": "LOG_UUID",
        "operation": "upsert",
        "version": 9,
        "payload": {
          "log_date": "2026-05-28",
          "value": "1500.000"
        }
      }
    ],
    "next_cursor": "OPAQUE_CURSOR_1051",
    "has_more": false
  },
  "meta": {
    "server_time": "2026-05-28T17:22:00Z",
    "api_version": "v1",
    "request_id": "REQUEST_UUID"
  }
}
```

State machine and bounded batches

queued → sending → accepted (durable ack), needs_review (409) or rejected (422). 429/5xx/network failure returns to queued with jittered exponential backoff. Push max100 mutations; pull default200/max500 changes. /sync is paginated because a long offline delta may exceed one safe response, superseding the initial unpaginated suggestion. Batch partial success is allowed: each mutation commits independently and returns accepted/conflict/rejected/dependency_pending. HTTP200 alone never means all accepted.

Cursor durability and pending projections

In one SQLite transaction: persist ack, canonical changes, rebased local projection and next_cursor. Crash before commit replays safely. Reapply unsent commuting deltas to the canonical value; keep absolute drafts in needs_review. Clear pending only after durable local apply. Drain has_more before saying synced. Domain writes, ledgers and server change rows commit together. Pull sequence is server-ordered, never based on client event clocks.

Retention and full bootstrap

Change journal/tombstones retained180 days; mutation receipts retained for account lifetime or equivalent non-replayable archive. 410 cursor_expired requires paginated bootstrap snapshot at snapshot_seq, then delta from that sequence. Preserve unsent outbox and review deleted/archived dependencies. Snapshot is consistent; new rows arrive through delta. Logging out / switching accounts never shares local queues across owners.

Timezones, schedules & streak freezes

The deterministic business contract. Current unfinished periods are pending, not failed. Frozen periods are visibly distinct from actual check-ins.

Day resolution and delayed offline events

Normal submission has immutable occurred_at UTC, captured IANA zone and date hint. Resolve DATE using trusted timezone history effective at occurred_at, validate context and return canonical date. Receipt time is never the habit date. Offline occurrences may be up to90 days old; future tolerance5 minutes. Event times are not hardware-attested: bounds and server-derived streaks do not make offline timing tamper-proof.

Midnight / travel example

```
Event:2026-05-29T06:30:00Z
Effective zone:America/Los_Angeles (PDT in this example)
Resolved local date:2026-05-28 at23:30, not UTC date29.
Changing profile zone to Europe/Paris on30 May does not rewrite this event.
Explicit backdate mode uses a separately validated local DATE within30 days and the habit active range; it does not invent an event timestamp.
```

Period and schedule rules

Resource / scenario	Method / rule	Contract / expected behavior
Daily / weekdays	Daily date or ISO weekdays1–7	Only dates in active ranges from start onward; effective definition governs target completion.
Interval	(local_date − anchor_date) mod N =0	Count scheduled periods, not intervening days. UI streak unit explicitly says scheduled periods.
Weekly count	Monday00:00 → next Monday local	Distinct completed dates. Target3 means3 days, not3 check-ins on1 day. Open week stays pending.
Current / longest streak	Eligible success/protected suffix	Closed suffix plus current completed daily period. An unfinished current period leaves prior streak intact. Weekly streak increments at closure.
Archive / restore / definition edit	Versioned boundaries	Archive pauses eligibility; restore starts today. Targets and schedules change next eligible period, never reinterpret old logs.

Freeze close-period transaction

At local close, evaluate completion, then order failed periods by end time and habit UUID. Lock user wallet; check effective opt-in policy, balance and unique habit/period usage. Debit1, insert usage, journal changes and recompute streak atomically. A whole failed week costs1, not1 per day. Open weeks consume nothing. Freeze preserves continuity but never counts as completion, awards XP or raises consistency.

Wallet and reconciliation

Resource / scenario	Method / rule	Contract / expected behavior
Policy history	user_freeze_policy_versions	id UUID; user_id UUID; enabled boolean; effective_at UTC. Index(user_id,effective_at). Opt-in is prospective, not retroactive to earlier misses.
Monthly grant	Cap2; unique user+month source	Start of local month credits max(0,2−balance) once. No stock above2 or indefinite accumulation.
Late real completion	Unique usage reconciliation	Mark protected period completed; refund min(1,2−balance) once; recompute streak/XP.
Refund at cap	Zero-delta audit event	No overflow stock banked. cap_reached reason, balance remains2; reconciliation final and explained to user.
Late edit removes completion	Reevaluate affected closed period	Active usage still protects. A refunded usage cannot reactivate: undoing that completion makes the period missed. A never-used eligible period may spend once.
Mobile example	Opt-in enabled20 May	Earlier misses4/9/13/16 not retroactively protected.20 May spends1 of2;1 remains.17–28 May streak12 includes protected20 May.

DST, invalid timezone context and backdating

Use IANA rules, never fixed numeric UTC offset. Spring reminder gaps move to next valid instant; fall-back repeat fires once using local-date+reminder key. Conflicting offline zone hint returns timezone_context_mismatch for review, never silently changes date. Explicit backdate rejects future/not-active dates and is limited to30 days, except verified own-backup restore. Habit starts may be backfilled up to30 days through declared backdate mode. Server never accepts claimed XP or streaks.

Rewards, reminders & data ownership

Transparent rewards and neutral reminders support the routine. Personal history stays private, portable and reconstructible from authoritative records.

One completion entitlement, not unlimited XP

Each valid completed habit-day has one10 XP entitlement. Binary set and threshold-reaching quantities/durations share the rule. Partial progress awards0; repeated increments after completion award0. Weekly habits may earn10 per distinct completed day, max one/day, without a separate weekly bonus. Reversal appends−10 and marks entitlement reversed; recompletion appends+10 for the next revision, so repeated toggling has zero net gain. Archive preserves valid past XP.

Published formula and consistent examples

```
level = 1 + floor(valid_xp / 100)
progress_into_level = valid_xp mod 100
next_level_threshold = 100 × level
240 XP → Level3,40 /100 toward4.
1,240 XP → Level13,40 /100 toward14.
Meditation +10 →1,250 XP, Level13,50 /100 toward14.
No streak, freeze or level value is accepted from a client.
```

Reminder delivery behavior

Resource / scenario	Method / rule	Contract / expected behavior
Local reminders	flutter_local_notifications	Per reminder definition and selected timezone mode. Reschedule on edits, archive/restore, approved zone changes and platform-required reboot handling.
Permission denied	Show device truth	Enabled definition is not guaranteed OS delivery. Offer Settings and next-due view; no guilt or repeated nagging.
Notification identity	Device + reminder UUID + local date/time slot	Replace rather than duplicate pending alarms; DST repeat fires once. Stable platform IDs and rolling schedule horizon.
Background execution	Best effort, not guaranteed	iOS/Android policies, battery saving and permissions can delay work. Logging stays local; foreground sync repairs backlog.
Server push	FCM/APNs milestone outbox	Opt-in and separately deduplicated. Never the daily reminder dependency; no loss-threat or guilt-driven copy.

Export jobs and portability backup

POST /exports queues an owner-bound snapshot: schema_version, generated_at UTC, snapshot_seq, habits/definition versions/active ranges, local dated logs and audited reward/freeze history. JSON is the portable backup; CSV is flat check-in history. Exclude password hashes, auth tokens, device push tokens and sensitive diagnostics. Encrypt private storage; signed links expire after24 hours and require authenticated ownership before issuance. Job failures are visible and never mutate history.

Safe restore / import

Stage upload, validate file size/schema/hash/ownership, then preview upserts and conflicts. Explicitly confirm before apply. Verified manifests from the same owner may restore older own history; untrusted CSV follows ordinary30-day backdate bounds. Remap foreign IDs; never trust embedded owner IDs or XP/freezes/streaks. Dedupe by habit identity+date, version-check updates, recompute entitlement effects. Preserve unsent work, audit per-item results and prevent hidden partial restores.

Security and deletion contract

Resource / scenario	Method / rule	Contract / expected behavior
Authorization	Owner scope on every lookup	UUIDs are not permission checks. Return owner-filtered404; test every route and sync mutation entity.
Credentials	Sanctum + secure device store	HTTPS; hashed passwords; keychain/keystore token storage; encrypted database where risk requires. No secrets in logs or exports.
Rate / replay controls	Per-user/device/IP limits	Auth throttling, payload caps, bounded sync, Retry-After, immutable mutation receipts and push-token revocation.
Account deletion	Revoke now; purge job	Recent reauthentication. Queue owned-data purge within30 days; private backups age out within90 days unless legal hold. Disclose retention policy.
Deletion status	Expiring status-only credential	Issue receipt on DELETE /me because full sessions revoke immediately. Receipt can only read deletion-job state.
Privacy default	No social broadcast	No public streaks, party penalties, guild damage, coercive notifications or cross-owner history.

Acceptance tests & release handoff

Release gates across services, SQL constraints, local SQLite transactions and real devices. Static Figma controls are not evidence that runtime behavior has been implemented.

Offline, concurrency and time boundaries

Resource / scenario	Method / rule	Contract / expected behavior
Three days offline	Three original local dates	Resolve original occurrences/context, not arrival day. Cursor advances only after durable local apply.
Duplicate +250 mL retry	Net +250 mL once	Same UUID/payload returns stored ack. Changed payload with same key returns409.
Concurrent increments	1,250 +250 +250 =1,750 mL	Each receipt atomic; no lost update. Repeated threshold crossings still have one XP entitlement.
Stale absolute edit	Version8 vs server9 →409	No overwrite. Preserve local1.50L draft, show server1.75L and reviewed resolution.
Delete vs queued increment	409 resource_deleted	No silent resurrection; explicit restore needs tombstone base_version.
Interrupted pull	Replay safe	Crash before commit leaves cursor unchanged. Retry duplicates are no-ops.
Expired180-day cursor	410 + preserved outbox	Consistent bootstrap snapshot, reconcile deletion, then delta; unsent logs not discarded.
29 May06:30 UTC in Pacific	28 May23:30 local	Trusted timezone history beats UTC date, arrival day and unchecked date hint.
DST gap/repeat	One intended reminder	Gap shifts forward; repeated fall-back clocktime fires once; future alarms reschedule on zone changes.

Streak, wallet and reward math

Resource / scenario	Method / rule	Contract / expected behavior
Weekly target3	Distinct completed days	3 days succeeds;3 taps on1 day does not. Open week pending; closed failed week costs at most1 freeze.
Two failures, one token	Exactly one protected period	Wallet lock + deterministic end-time/UUID order; no negative balance or double spend.
Monthly refill retry	Cap2	Unique month source key; concurrent jobs do not duplicate credit.
Late protected completion	One cap-aware refund	Complete replaces protected, usage reconciled once. Full balance2 gets visible zero-credit cap audit.
Archive then restore	No archived-gap misses	History intact; new range today; target changes do not reinterpret older values.
Complete/reverse/recomplete	+10 −10 +10 = net10	Unique entitlement revisions; partial quantity and freeze-only status earn0.
Level boundaries	99→1;100→2;1,250→13	Progress0…99, next threshold100×level; API math matches mobile display.
Monthly totals	(37+22+21+15)/(54+27+25+19)	95/125 =76%. Category values sum; frozen periods are not successes.
Weekly closed-period total	3+2+4 =9 completions	9/15 =60% through27 May;90 XP. Current28 May remains open and outside that denominator.
Heatmap streak	17–28 May =12 eligible days	20 May protected,16 May missed; current12 includes11 successes and1 protected day.

Privacy, permission and UX checks

Resource / scenario	Method / rule	Contract / expected behavior
Permission denied	Honest delivery state	Definition retained, Settings route visible; offline logging unaffected.
Cross-owner IDs	404, no leaked values	Habits/logs/reminders/devices/exports/imports/deletion receipts and every sync entity.
Export snapshot	Dates/counts preserved	JSON own-manifest round trip; CSV escaping; UTC versus local dates; secret exclusion; expiring owner-only URLs.
Bad import / restore conflict	Preview or validation error	No claimed-ledger XP minting or unreviewed overwrite; visible per-item result.
Native accessibility	Semantics + scalable text	Screen-reader labels; non-color-only statuses; heatmap date/status labels; ≥44 logical px hit targets; support dynamic type and safe areas.
Empty / error / offline	User intent preserved	First-habit route, zero-duration validation, durable outbox and conflict review retain the rest of the user's input.

Definition of done

Owner isolation and migration rollbacks pass. Durable idempotency covers every mutation. UTC event time and local DATE are distinct. Sync converges after retries/offline/multi-device use. Streak/freeze/XP races pass. Local notifications tested on real iOS/Android with denied permissions and battery restrictions. Export/restore/deletion verified. No functionality is claimed merely because it appears in Figma.